

# Zero-Shot Cross-Lingual Transfer for Code-Mixed Conversation Summarization

Akira Nair | Aneesh Edara | Prednya Ramesh | Ricky Raup

## Extension: Zero-Shot Transfer to Unseen Language Pairs:

**Motivation:** Code-mixed conversations (e.g., Hinglish, Spanglish) are prevalent in multilingual communities, but building summarization systems for each of the 2,450+ possible language pairs is impractical. We investigate whether a model trained on a subset of code-mixed pairs can generalize to completely unseen pairs without any target-language training data.

**Approach:** We train models on three code-mixed language pairs from CS-SUM dataset (Chinese-English, Tamil-English, Malay-English; 1,824 training examples) and evaluate zero-shot transfer to three unseen pairs (Japanese-English, Korean-English, Hindi-English). We compare four architectures:

| Model               | Type     | Params | Training |
|---------------------|----------|--------|----------|
| mT5-base            | Enc-Dec  | 580M   | Full     |
| mBART-50            | Enc-Dec  | 611M   | Full     |
| Llama-3.2-3B        | Dec-Only | 3B     | LoRA     |
| <b>Llama-3.1-8B</b> | Dec-Only | 8B     | LoRA     |

## Results:

Table 1: ROUGE-L Performance by Language Pair (\* = Zero-shot)

| Model           | Zh-En        | Ta-En        | Ms-En        | Ja-En*       | Ko-En* | Hi-En*       |
|-----------------|--------------|--------------|--------------|--------------|--------|--------------|
| mT5             | 0.061        | 0.047        | 0.068        | 0.000        | 0.051  | 0.167        |
| mBART           | 0.333        | 0.293        | 0.337        | <b>0.541</b> | 0.228  | 0.210        |
| Llama-3B        | 0.303        | 0.302        | 0.287        | 0.440        | 0.390  | <b>0.481</b> |
| <b>Llama-8B</b> | <b>0.384</b> | <b>0.336</b> | <b>0.345</b> | <b>0.446</b> | 0.355  | 0.298        |

Table 2: Transfer Rate Analysis

| Model               | In-Domain    | Zero-Shot    | Transfer Rate |
|---------------------|--------------|--------------|---------------|
| mT5-base            | 0.059        | 0.073        | 124% (failed) |
| mBART-50            | 0.321        | 0.326        | 102%          |
| <b>Llama-3.2-3B</b> | 0.297        | <b>0.437</b> | <b>147%</b>   |
| <b>Llama-3.1-8B</b> | <b>0.355</b> | 0.366        | <b>103%</b>   |

## Key Findings:

- Zero-shot transfer works:** All Llama models achieve >100% transfer rate, performing as well or better on unseen languages than trained ones.
- Decoder-only > Encoder-decoder:** Llama models significantly outperform mBART/mT5 on zero-shot transfer, suggesting LLMs' multilingual pretraining enables better generalization.
- Surprising result:** Llama-3.2-3B achieves 147% transfer rate - zero-shot performance (0.437 ROUGE-L) exceeds in-domain performance (0.297), indicating the model learns language-agnostic summarization patterns.
- Scale provides stability:** Llama-3.1-8B shows more consistent performance across languages (0.298-0.446) compared to Llama-3.2-3B's higher variance (0.287-0.481).

## Connection to CoT Paper:

Our instruction-tuned Llama models implicitly leverage chain-of-thought reasoning patterns from pretraining. As shown in "Long-Chain-of-Thought Reasoning Across Languages", multilingual LLMs can transfer reasoning capabilities across languages. Our results support this finding - zero-shot transfer rates exceeding 100% suggest the model transfers summarization "reasoning" rather than language-specific patterns.

## Conclusion:

We demonstrate the first successful zero-shot cross-lingual transfer for code-mixed conversation summarization. A single model trained on 3 language pairs generalizes to unseen pairs with 100%+ transfer rates, enabling practical deployment across thousands of code-mixed language combinations without target-language data.